A uniform quadratic wedge for consecutive Toeplitz minors of the Riemann ξ-coefficients

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Abstract. Let $\frac{1}{8}\xi(\frac{1}{2}+\frac{ix}{2}) = \sum_{k\geq 0}(-1)^k a_k x^{2k}$, so that $a_k > 0$, and let $D_{r,k} = \det[a_{k+j-i}]_{i,j=0}^{r-1}$ be the consecutive Toeplitz minors of the coefficient sequence. The Riemann Hypothesis implies that (a_k) is a Pólya frequency sequence, hence $D_{r,k} \geq 0$ for all r, k. Michałowski recently proved unconditionally that $D_{r,k} > 0$ whenever $r \geq 2$ and $k \geq 10^{18} r^3$. We prove that

$$D_{r,k} > 0 \qquad \text{for all } r \geq 2 \text{ and } k \geq 10^{11} r^2.$$

The improvement comes from replacing the three-point log-quadratic comparison sequence by a four-point q-Pochhammer comparison sequence $c_s = (1-\alpha)^s/(\alpha;\mathfrak{q})_s$ that reproduces the normalized coefficients at four consecutive indices exactly. The reversed Hankel block of this reference is diagonally congruent to a matrix with an exact q-Pascal LDL^T factorization (a consequence of the q-Chu–Vandermonde summation) with alternating pivots, so the dilation/whitening argument applies to it unchanged, while the logarithmic mismatch now has four interpolation zeros. The weighted remainder therefore starts at fourth order and scales like r^2/k instead of $(r^3/k)^{1/2}$. The only analytic inputs are Michałowski's all-degree Taylor bound for $\log a$ and the Liu–Mao asymptotic for the log-curvature of (a_k); all remaining constants are certified in interval arithmetic.

Keywords. Riemann ξ-function; Toeplitz minors; Pólya frequency sequences; total positivity; q-Pochhammer symbol; q-Chu–Vandermonde; Sylvester inertia.

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1. Introduction

1.1. Setting

Let $\xi(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(s/2)\zeta(s)$ and write, following [10],

$$\Xi(x) = \frac{1}{8}\,\xi\Big(\frac{1}{2}+\frac{ix}{2}\Big) = \sum_{k\geq 0}(-1)^k a_k x^{2k}, \qquad a_k = \frac{1}{(2k)!}\int_0^\infty u^{2k}\Phi(u)\,du > 0, \tag{1.1}$$

where $\Phi(u) = \sum_{n\geq 1}(2\pi^2 n^4 e^{9u} - 3\pi n^2 e^{5u})\exp(-\pi n^2 e^{4u}) > 0$. For $r \geq 1$ and $k \geq 0$ set

$$D_{r,k} = \det\left[a_{k+j-i}\right]_{i,j=0}^{r-1} \qquad (a_\ell = 0 \text{ for } \ell < 0).$$

The function $G(z) = \sum_k a_k z^k$ is entire of order $1/2$ and $\Xi(x) = G(-x^2)$. The Riemann Hypothesis (RH) is equivalent to all zeros of G being real and negative, and by the

Aissen–Schoenberg–Whitney–Edrei theorem [1, 3] this happens if and only if (a_k) is a Pólya frequency sequence, i.e. every minor of the Toeplitz matrix $[a_{j-i}]_{i,j\geq 0}$ is nonnegative. The consecutive minors $D_{r,k}$ form a natural two-parameter subfamily; under RH they are all nonnegative.

Unconditional information on $D_{r,k}$ is of two kinds. For each fixed r, $D_{r,k} > 0$ for $k \geq N(r)$ with a non-explicit $N(r)$ (Katkova [8]); for $r = 2$ this is the Turán inequality [2]. Liu and Mao [9] recently proved that (a_k) (equivalently the sequence $\lambda_n = \xi^{(2n)}(\frac{1}{2})/(2n)!$) is infinitely log-concave. Uniformly in r, Michałowski [10] proved the explicit cubic wedge

$$D_{r,k} > 0 \qquad (r \geq 2,\ k \geq 10^{18} r^3). \tag{1.2}$$

Related but different in nature are the hyperbolicity results for Jensen polynomials of ξ [5, 6], which concern a different two-parameter family of determinantal conditions.

1.2. Result

Main Theorem. *For every integer $r \geq 2$ and every integer $k \geq 10^{11} r^2$ one has $D_{r,k} > 0$.*

The constant 10^{11} is not optimized; the argument closes for every $C > 1.35 \cdot 10^{10}$ (Remark 8.2). The theorem says nothing about the RH-critical regime $k = O(r)$.

1.3. Strategy

We follow the architecture of [10]: compare the reversed Hankel block $H = [a_{k+d-i-j}]_{i,j=0}^{d}$, $d = r - 1$, with an explicit reference block $M = SJS^T$ of known inertia, and show that the *whitened* perturbation $S^{-1}(H' - M)S^{-T}$ has spectral norm < 1, where H' is a diagonal rescaling of H; Sylvester's law then fixes the sign of $\det H$. The new ingredient is the reference. In [10] the reference is $c_s = q_k^{s(s-1)/2}$, which matches $\log a_{k+s}$ at the three points $s = -1, 0, 1$, so the logarithmic error h has a cubic Taylor term of size k^{-2}. Here we use the q-Pochhammer sequence

$$c_s = \frac{(1-\alpha)^s}{(\alpha;\mathfrak{q})_s},$$

whose two parameters are chosen so that c matches the normalized coefficients at $s = -1, 0, 1, 2$. Two facts make this work:

1. the reversed Hankel block of c is diagonally congruent to the matrix

$$U_{ij} = \frac{(B;Q)_{i+j}}{(B;Q)_i\,(B;Q)_j},$$

   which has an exact LDL^T factorization with the *same* q-Pascal factor $L = [\begin{smallmatrix} i \\ j \end{smallmatrix}]_Q$ as the q-Vandermonde matrix and with pivots of alternating sign (Proposition 5.2);

2. the response of the whitened block to a diagonal dilation depends only on L and on the pivot ratios, and the new pivot ratios give a whitened sub-diagonal that is no larger than before (Lemma 6.2).

Since the error h now vanishes at four nodes, it equals $\sum_{n\geq 4} \beta_n (s^n - \mathcal{I}_3[s^n])$ with $|\beta_n| \lesssim 40^n k^{1-n}$, and the relevant weight is $w \asymp \sqrt{rk}$; the leading term is $k^{-3} w^4 \asymp r^2/k$.

1.4. Inputs and logical dependencies

We use exactly two analytic results from the literature, stated in Section 2: the k-unit form of Michałowski's all-degree bound ([10, Prop. 3.7, first display]) and the Liu–Mao estimate [9, Lemma 3.3]. We do *not* use [10, Lemma 2.1] (the bounds $1/(2k) < \tau_k < 4/k$) nor its τ-unit form of Gate A: all information on τ_k comes from Input B. The dilation estimates and the inertia lemma are re-proved here for the new reference. All scalar inequalities are certified with Arb [7]; scripts are provided as ancillary files.

Notation

$(a;q)_n = \prod_{j=0}^{n-1}(1-aq^j)$ for $n \geq 0$, $(a;q)_{-m} = 1/(aq^{-m};q)_m$ for $m > 0$, and $\begin{bmatrix} n \\ m \end{bmatrix}_Q = (Q;Q)_n/((Q;Q)_m(Q;Q)_{n-m})$. Throughout $r \geq 2$, $d = r-1$, and $\|\cdot\|$ is the spectral norm.

2. Analytic inputs

Put $f = \log a$, where $a(z) = I(z)/\Gamma(2z+1)$ and $I(z) = \int_0^\infty u^{2z}\Phi(u)\,du$ is holomorphic on $\operatorname{Re} z > -\frac{1}{2}$; thus $a(n) = a_n$ for integers $n \geq 0$.

Input A (Michałowski, Gate A) ([10, Lemma 3.6, Prop. 3.7])**.** *For every integer $k \geq 10^9$, f is holomorphic on $|z-k| \leq 0.05k$ (with $f(n) = \log a_n$ at integers), and*

$$\Big|\frac{f^{(n)}(k)}{n!}\Big| \leq 3 \cdot 40^n\, k^{1-n} \qquad (n \geq 3). \tag{2.1}$$

Define the log-curvature

$$q_k = \frac{a_{k-1}a_{k+1}}{a_k^2} = e^{-\tau_k}, \qquad \tau_k > 0.$$

Liu and Mao use $F(x) = \xi(\frac{1}{2} + \sqrt{x}) = \sum \lambda_n x^n$. Since $\frac{1}{8}\xi(\frac{1}{2} + \frac{\sqrt{z}}{2}) = \sum a_n z^n = \frac{1}{8}F(z/4)$, we have $a_n = \lambda_n/(8 \cdot 4^n)$, and therefore $\lambda_{n-1}\lambda_{n+1}/\lambda_n^2 = q_n$. Let W be the principal branch of the Lambert function and

$$\mathcal{T}(z) = \frac{2\,W(2z/\pi)}{z\,(1+W(2z/\pi))}.$$

Input B (Liu–Mao, Lemma 3.3) ([9, Lemma 3.3, (3.53)])**.** *For real $n \geq 12000$ there is a function g_0, holomorphic on $D_0 = \{z : |z-n| \leq n/2\}$ and real on $D_0 \cap \mathbb{R}$, with $g_0(m) = \log(\lambda_m^2/\lambda_{m-1}\lambda_{m+1}) = \tau_m$ for integers $m \in D_0$, and*

$$|g_0(z) - \mathcal{T}(z)| \leq \frac{20}{n^2} \qquad (z \in D_0).$$

Remark 2.1. Input A is used only through the Taylor coefficients of f at an integer centre; any analytic interpolation of $\log a_n$ with these bounds would do, and it need not coincide with the interpolation g_0 of Input B, since only the values at integers enter the comparison (Section 7).

3. Curvature windows

Lemma 3.1. *For every integer $k \geq 12000$,*

$$1.71 < \frac{12}{7} - \frac{20}{k} \leq k\tau_k \leq 2 + \frac{20}{k} < 2.002.$$

Proof. Write $w = W(2x/\pi)$. Since $6e^6 < 2 \cdot 12000/\pi$ and W is increasing, $w > 6$ for $x \geq 12000$, so $x\mathcal{T}(x) = 2w/(1+w) \in [12/7, 2)$. Apply Input B with $n = k$ at $z = k$. □

Lemma 3.2. *For every integer $k \geq 12000$, $\Delta_k := \tau_k - \tau_{k+1}$ satisfies*

$$1.675 < \frac{576}{343}\Big(\frac{k}{k+1}\Big)^2 - \frac{40}{k-2} \leq k^2\Delta_k \leq 2 + \frac{40}{k-2} < 2.004.$$

In particular $q_k < q_{k+1}$.

Proof. With $w = W(2x/\pi)$ and $w'(x) = w/(x(1+w))$ one computes

$$-\mathcal{T}'(x) = \frac{2}{x^2}\, g(w), \qquad g(w) = \frac{w^2(w+2)}{(1+w)^3}.$$

Since $(\log g)'(w) = \frac{2}{w} + \frac{1}{w+2} - \frac{3}{w+1} = \frac{w+4}{w(w+1)(w+2)} > 0$, g increases from $g(6) = 288/343$ to 1; hence $\frac{576}{343}x^{-2} \leq -\mathcal{T}'(x) < 2x^{-2}$ for $x \geq 12000$, and $\mathcal{T}(k) - \mathcal{T}(k+1) \in [\frac{576}{343}(k+1)^{-2},\, 2k^{-2}]$. Let $E = g_0 - \mathcal{T}$ with $n = k$ in Input B. For $\eta \in [k, k+1]$ the disc $|z - \eta| \leq k/2 - 1$ lies in D_0, so Cauchy's estimate gives $|E'(\eta)| \leq (20/k^2)/(k/2-1) = 40/(k^2(k-2))$, whence $|E(k) - E(k+1)| \leq 40/(k^2(k-2))$. Both bounds are monotone in k in the favourable direction; the numerical values at $k = 12000$ are certified in Appendix A. □

4. The four-point q-Pochhammer reference

Fix an integer $k \geq 12000$ and normalize

$$\rho = \frac{a_{k+1}}{a_k}, \qquad b_s = \frac{a_{k+s}}{a_k\, \rho^s},$$

so that $b_{-1} = q_k =: u$, $b_0 = b_1 = 1$, $b_2 = q_{k+1} =: v$, and $0 < u < v < 1$ by Lemmas 3.1–3.2.

Definition 4.1. Let

$$\mathfrak{q} = \frac{1-v}{v(1-u)}, \qquad \alpha = \frac{(1-u)(1-v)}{v-u}, \qquad \theta = \log \mathfrak{q}, \qquad c_s = \frac{(1-\alpha)^s}{(\alpha;\mathfrak{q})_s} \quad (s \in \mathbb{Z}).$$

Lemma 4.2. *$c_{-1} = u$, $c_0 = c_1 = 1$, $c_2 = v$. Moreover, with $\sigma = \tau_{k+1}$ and $\Delta = \tau_k - \sigma$,*

$$\alpha = \frac{2\sinh(\tau_k/2)\sinh(\sigma/2)}{\sinh(\Delta/2)}, \qquad \theta = \frac{\tau_k + \sigma}{2} + \log\frac{\sinh(\sigma/2)}{\sinh(\tau_k/2)}.$$

Proof. $c_0 = 1$, $c_1 = 1$, $c_2 = (1-\alpha)/(1-\alpha\mathfrak{q})$ and $c_{-1} = (1-\alpha\mathfrak{q}^{-1})/(1-\alpha)$. Substituting the definitions of $\alpha, \mathfrak{q}$ gives $c_2 = v$ and $c_{-1} = u$ after clearing denominators (a polynomial identity in u, v). The hyperbolic forms follow from $1 - e^{-x} = 2e^{-x/2}\sinh(x/2)$. □

Lemma 4.3. *For $k \geq 12000$: $1.45 < \alpha < 2.40$ and $0.53 < k\theta < 2.002$. In particular $\mathfrak{q} > 1$.*

Proof. Using $x \leq \sinh x \leq xe^{x^2/6}$, α lies between $\frac{\tau_k\sigma}{\Delta}e^{-\Delta^2/24}$ and $\frac{\tau_k\sigma}{\Delta}e^{(\tau_k^2+\sigma^2)/24}$. For θ: $(\tau_k + \sigma)/2 = \sigma + \Delta/2 \geq \sigma$, $\log(\sigma/\tau_k) = -\log(1 + \Delta/\sigma) \geq -\Delta/\sigma$, and $\log\frac{\sinh(\sigma/2)}{\sinh(\tau_k/2)} \geq \log\frac{\sigma}{\tau_k} - \frac{\tau_k^2}{24}$; hence $\theta \geq \sigma - \Delta/\sigma - \tau_k^2/24$. Also $\theta < \tau_k$ because sinh is increasing. Insert Lemmas 3.1–3.2 and $k\sigma \geq k\tau_k - k\Delta_k$; see Appendix A. □

Lemma 4.4 (Positivity)**.** *If $(r-1)\theta < \log\alpha$, then $c_s > 0$ for $|s| \le r-1$, and every factor $1-\alpha\mathfrak{q}^j$ with $-(r-1) \le j \le r-2$ is negative.*

Proof. $\alpha\mathfrak{q}^j \ge \alpha e^{-(r-1)\theta} > 1$ for $j \ge -(r-1)$. For $s > 0$ numerator and denominator of c_s both have sign $(-1)^s$; for $s = -m < 0$, $c_s = (1-\alpha)^{-m}\prod_{j=1}^{m}(1-\alpha\mathfrak{q}^{-j})$ has $2m$ negative factors. □

Remark 4.5. The condition $1 < \mathfrak{q} < \alpha$ alone does *not* suffice for positivity at negative indices; the hypothesis $(r-1)\theta < \log\alpha$ is what is needed, and it holds in the wedge of the Main Theorem with enormous room (Appendix A).

4.1. Analytic continuation and Taylor coefficients

Put $z_0 = 1/\alpha \in (0.41, 0.69)$ and $Q = \mathfrak{q}^{-1} = e^{-\theta}$. Writing $1-\alpha\mathfrak{q}^j = -\alpha\mathfrak{q}^j(1-z_0Q^j)$ gives $c_s = (1-z_0)^s Q^{s(s-1)/2}/(z_0;Q)_s$ for integers s (for $s<0$ by the cocycle relation $(a;q)_{m+n} = (a;q)_m(aq^m;q)_n$, valid for all integers m,n). With $(z_0;Q)_s = (z_0;Q)_\infty/(z_0Q^s;Q)_\infty$ and $\log(x;Q)_\infty = -\sum_{m\ge1} x^m/(m(1-Q^m))$ we obtain, for $|s| < \theta^{-1}\log(1/z_0)$,

$$\ell(s) := s\log(1-z_0) - \frac{\theta}{2}s(s-1) + \sum_{m\ge1}\frac{z_0^m(1-e^{-m\theta s})}{m\,(1-e^{-m\theta})}, \tag{4.1}$$

a holomorphic function, real on the real axis, with $e^{\ell(s)} = c_s$ at integers s in its domain.

Lemma 4.6. *For $n \ge 3$, $|[s^n]\,\ell(s)| \le \kappa^n/n + \kappa^{n-1}/(n(n-1))$ with $\kappa = \theta/(1-z_0)$. For $k \ge 12000$ this gives $|[s^n]\ell(s)| \le (9/k)^{n-1}$.*

Proof. For $n \ge 3$ only the series contributes, and $[s^n](1-e^{-m\theta s}) = -(-m\theta)^n/n!$, so

$$|[s^n]\ell| \le \frac{\theta^n}{n!}\sum_{m\ge1}\frac{m^{n-1}z_0^m}{1-e^{-m\theta}} \le \frac{\theta^n}{n!}\Big(\sum_m m^{n-1}z_0^m + \frac{1}{\theta}\sum_m m^{n-2}z_0^m\Big),$$

using $1/(1-e^{-x}) \le 1 + 1/x$. Since $\sum_{m\ge1} m^p z^m = zA_p(z)/(1-z)^{p+1}$ with the Eulerian polynomial A_p (nonnegative coefficients, $A_p(1) = p!$), $\sum_m m^p z^m \le p!/(1-z)^{p+1}$ for $0 < z < 1$, $p \ge 0$. This gives the first bound. By Lemma 4.3, $k\kappa = k\theta\,\alpha/(\alpha-1) < 2.002\cdot 1.45/0.45 < 6.46 < 8.4$, and $8.4^n/(nk) + 8.4^{n-1}/(n(n-1)) < 9^{n-1}$ for $n \ge 3$, $k \ge 12000$. □

5. The reversed Hankel block and its q-Pascal factorization

Set $d = r-1$ and

$$M_{ij} = c_{d-i-j}, \qquad 0 \le i,j \le d, \qquad Q = \mathfrak{q}^{-1}, \qquad B = \alpha\mathfrak{q}^{d-1}.$$

Lemma 5.1 (Congruence)**.** *$M = c_d\,A\,U\,A$ with $A = \mathrm{diag}(A_i)$, $A_i = (B;Q)_i/(1-\alpha)^i$, and*

$$U_{ij} = \frac{(B;Q)_{i+j}}{(B;Q)_i\,(B;Q)_j}.$$

Under the hypothesis of Lemma 4.4, $c_d > 0$ and $A_i > 0$.

Proof. By the integer cocycle, $(\alpha;\mathfrak{q})_d = (\alpha;\mathfrak{q})_{d-n}(\alpha\mathfrak{q}^{d-n};\mathfrak{q})_n$ for every $0 \le n \le 2d$, including $d-n<0$, and $(\alpha\mathfrak{q}^{d-n};\mathfrak{q})_n = \prod_{l=0}^{n-1}(1-\alpha\mathfrak{q}^{d-1-l}) = (B;Q)_n$. Hence $c_{d-n} = c_d\,(B;Q)_n/(1-\alpha)^n$, which is the claim with $n=i+j$. Signs follow from Lemma 4.4: $1-BQ^l = 1-\alpha\mathfrak{q}^{d-1-l} < 0$ for $0 \le l \le d-1$, so $(B;Q)_i$ and $(1-\alpha)^i$ both have sign $(-1)^i$. □

Proposition 5.2 (q-Pascal LDL^T)**.** *For indeterminates B, Q and all $i,j \ge 0$,*

$$U_{ij} = \sum_{m=0}^{\min(i,j)} \begin{bmatrix} i \\ m \end{bmatrix}_Q \begin{bmatrix} j \\ m \end{bmatrix}_Q D_m, \qquad D_m = B^m Q^{m(m-1)} \frac{(Q;Q)_m}{(B;Q)_m}.$$

Thus $U = LDL^T$ with $L_{ij} = \begin{bmatrix} i \\ j \end{bmatrix}_Q$ and $D = \operatorname{diag}(D_m)$. In our situation $0<Q<1$ and $BQ^{m-1} = \alpha\mathfrak{q}^{d-m} > 1$ for $1 \le m \le d$, so $\operatorname{sgn} D_m = (-1)^m$.

Proof. From $(Q;Q)_i/(Q;Q)_{i-m} = (-1)^m Q^{im-\binom{m}{2}}(Q^{-i};Q)_m$ we get

$$\begin{bmatrix} i \\ m \end{bmatrix}_Q \begin{bmatrix} j \\ m \end{bmatrix}_Q D_m = \frac{(Q^{-i};Q)_m (Q^{-j};Q)_m}{(Q;Q)_m (B;Q)_m} \, (BQ^{i+j})^m.$$

Hence the sum equals ${}_2\phi_1(Q^{-i}, Q^{-j}; B; Q, BQ^{i+j})$, which by the q-Chu–Vandermonde summation ${}_2\phi_1(q^{-n}, b; c; q, cq^n/b) = (c/b;q)_n/(c;q)_n$ [4, §1.5 and Appendix II], with $n=i$, $b=Q^{-j}$, $c=B$, equals $(BQ^j;Q)_i/(B;Q)_i = U_{ij}$. The sign of D_m: $(Q;Q)_m > 0$ and every factor $1-BQ^l$, $0 \le l \le m-1$, of $(B;Q)_m$ is negative. □

Consequently

$$M = SJS^T, \qquad S = \sqrt{c_d}\, A\, L\, |D|^{1/2}, \qquad J = \operatorname{diag}\left((-1)^m\right)_{m=0}^{d}. \tag{5.1}$$

6. Dilation response of the whitened reference

For a function ϕ defined at the integers $|s| \le d$ write $M \circ \phi := [c_{d-i-j}\phi(d-i-j)]_{i,j}$ and define the *response*

$$\mathcal{R}(\phi) = \|S^{-1}(M \circ \phi)S^{-T}\|.$$

Since $M \circ \phi = c_d A(U \circ \phi)A$ and diagonal matrices commute with the Hadamard product structure, $\mathcal{R}(\phi) = \||D|^{-1/2}L^{-1}(U\circ\phi)L^{-T}|D|^{-1/2}\|$; in particular A and c_d play no role.

Lemma 6.1. *Let $\Lambda_\beta = \operatorname{diag}(Q^{\beta i})_{i=0}^d$ and $R_\beta = L^{-1}\Lambda_\beta L$. Then R_1 is lower bidiagonal with*

$$(R_1)_{mm} = Q^m, \qquad (R_1)_{m,m-1} = -Q^{m-1}(1-Q^m).$$

Proof. We need $\Lambda_1 L = L R_1$, i.e. $Q^i \begin{bmatrix} i \\ j \end{bmatrix}_Q = Q^j \begin{bmatrix} i \\ j \end{bmatrix}_Q - Q^j(1-Q^{j+1}) \begin{bmatrix} i \\ j+1 \end{bmatrix}_Q$. This follows from the q-Pascal relation $(1-Q^{j+1})\begin{bmatrix} i \\ j+1 \end{bmatrix}_Q = (1-Q^{i-j})\begin{bmatrix} i \\ j \end{bmatrix}_Q$. □

Lemma 6.2 (Whitened sub-diagonal)**.** *Let $\widehat{R}_\beta = |D|^{-1/2} R_\beta |D|^{1/2}$. Then $(\widehat{R}_1)_{mm} = Q^m$ and*

$$|(\widehat{R}_1)_{m,m-1}| = \sqrt{(1-Q^m)(Q^{m-1}-B^{-1})} \le \sqrt{1-Q^m} \le \sqrt{m\theta}.$$

Proof. By Proposition 5.2,

$$\frac{D_m}{D_{m-1}} = -\frac{BQ^{2m-2}(1-Q^m)}{BQ^{m-1}-1}, \qquad (\widehat{R}_1)_{m,m-1} = (R_1)_{m,m-1}\,\frac{|D_{m-1}|^{1/2}}{|D_m|^{1/2}}.$$

Squaring and simplifying gives the identity; then use $0 < Q^{m-1} - B^{-1} < 1$ and $1 - e^{-m\theta} \le m\theta$. □

Remark 6.3. For the q-Vandermonde reference of [10] the corresponding quantity is $\sqrt{1-Q^m}$; the new reference multiplies it by $\sqrt{Q^{m-1}-B^{-1}} \approx \sqrt{1-1/\alpha} < 1$. Nothing else in the dilation argument depends on the pivots.

Lemma 6.4 (Generator)**.** *Let $x = r\theta \le 1/64$. Then $N := \widehat{R}_1 - I$ satisfies $\|N\| \le x + \sqrt{x} \le 9/64$, the principal logarithm $G = \log \widehat{R}_1 = \sum_{j\ge1}(-1)^{j+1}N^j/j$ satisfies $\|G\| \le \frac{72}{55}\sqrt{x}$, $\widehat{R}_\beta = e^{\beta G}$ for all real β, and*

$$\theta d + 2\|G\| \le 4\sqrt{r\theta} =: t.$$

Proof. The diagonal part of N has entries $|Q^m - 1| \le m\theta \le x$ and its sub-diagonal part has entries $\le \sqrt{x}$ (Lemma 6.2); a diagonal and a weighted shift have norms equal to their largest entries, so $\|N\| \le x + \sqrt{x}$. Then $\|G\| \le -\log(1-\|N\|) \le \|N\|/(1-\|N\|) \le \frac{64}{55}\cdot\frac{9}{8}\sqrt{x}$. Next, $R_\beta = \exp(\beta L^{-1}\Lambda' L)$ with $\Lambda' = \mathrm{diag}(i\log Q)$, so $\widehat{R}_\beta = e^{\beta\widehat{G}}$ with $\widehat{G} = |D|^{-1/2}L^{-1}\Lambda' L|D|^{1/2}$. Both $\widehat{G}$ and G equal $P(\widehat{R}_1)$, where P is the Lagrange polynomial with $P(Q^i) = i\log Q$ (the eigenvalues Q^i of $\widehat{R}_1$ are distinct and $\widehat{R}_1$ is diagonalizable); hence $G = \widehat{G}$. Finally $\theta d + 2\|G\| \le x + \frac{144}{55}\sqrt{x} \le (\frac{1}{8} + \frac{144}{55})\sqrt{x} < 4\sqrt{x}$. □

Proposition 6.5 (All-degree response)**.** *If $r\theta \le 1/64$, then $\mathcal{R}((\theta s)^n) \le t^n$ for every integer $n \ge 0$, with $t = 4\sqrt{r\theta}$.*

Proof. For $s = d - i - j$ we have $e^{\beta\theta s} = e^{\beta\theta d}Q^{\beta i}Q^{\beta j}$, so $M \circ e^{\beta\theta s} = e^{\beta\theta d}\Lambda_\beta M \Lambda_\beta$. Since A commutes with Λ_β, $S^{-1}\Lambda_\beta S = |D|^{-1/2}L^{-1}\Lambda_\beta L|D|^{1/2} = \widehat{R}_\beta = e^{\beta G}$, and therefore

$$F(\beta) := S^{-1}(M \circ e^{\beta\theta s})S^{-T} = e^{\beta\theta d}\, e^{\beta G}\, J\, e^{\beta G^T}.$$

Entrywise, $F^{(n)}(0) = S^{-1}(M \circ (\theta s)^n)S^{-T}$. Expanding the three exponentials, $F^{(n)}(0) = \sum_{a+b+c=n} \frac{n!}{a!\,b!\,c!}(\theta d)^a G^b J (G^T)^c$, whose norm is at most $(\theta d + 2\|G\|)^n \le t^n$. □

Definition 6.6 (Weighted algebra)**.** For $w \ge 1$ let $\mathcal{A}_w$ be the space of power series $\phi(s) = \sum_{n\ge0}\phi_n s^n$ with $\|\phi\|_{\mathcal{A}} = \sum_n |\phi_n| w^n < \infty$. It is a commutative Banach algebra, and $\|e^\phi - 1\|_{\mathcal{A}} \le e^{\|\phi\|_{\mathcal{A}}} - 1$.

Corollary 6.7. *Let $r\theta \le 1/64$ and $w = t/\theta = 4\sqrt{r/\theta}$. Then $w \ge 32r > d$, and for every $\phi \in \mathcal{A}_w$, $\mathcal{R}(\phi) \le \|\phi\|_{\mathcal{A}}$.*

Proof. $w/r = 4/\sqrt{r\theta} \ge 32$. The series $\phi(s) = \sum \phi_n s^n$ converges absolutely at every integer $|s| \le d$, so $M \circ \phi = \sum_n \phi_n\, M \circ s^n$ entrywise, and $\mathcal{R}(s^n) = \theta^{-n}\mathcal{R}((\theta s)^n) \le w^n$. □

7. The four-zero remainder

Let $k \geq 10^9$ and $r\theta \leq 1/64$. By Input A, $\log b_s = f(k+s) - f(k) - s\,(f(k+1) - f(k))$ at integers $|s| \leq d$, and the right-hand side is holomorphic for $|s| < 0.05k$. Define

$$h(s) = f(k+s) - f(k) - s\,(f(k+1) - f(k)) - \ell(s), \qquad \beta_n = [s^n]h(s).$$

Then $b_s = c_s\, e^{h(s)}$ at the integers $|s| \leq d$, and, by Lemma 4.2,

$$h(-1) = h(0) = h(1) = h(2) = 0.$$

Lemma 7.1. *For $n \geq 3$, $|\beta_n| \leq 3 \cdot 40^n k^{1-n} + (9/k)^{n-1} \leq 4 \cdot 40^n k^{1-n}$.*

Proof. For $n \geq 2$, $\beta_n = f^{(n)}(k)/n! - [s^n]\ell$; use (2.1), Lemma 4.6 and $9^{n-1} \leq 40^n$. □

Let $\mathcal{I}_3$ denote Lagrange interpolation at the nodes $-1, 0, 1, 2$, with basis

$$\ell_{-1}(s) = -\tfrac{s^3}{6} + \tfrac{s^2}{2} - \tfrac{s}{3}, \quad \ell_0(s) = \tfrac{(s+1)(s-1)(s-2)}{2}, \quad \ell_1(s) = -\tfrac{s^3}{2} + \tfrac{s^2}{2} + s, \quad \ell_2(s) = \tfrac{s^3}{6} - \tfrac{s}{6},$$

and put $P_n(s) = s^n - \mathcal{I}_3[s^n]$; note $P_n = 0$ for $n \leq 3$.

Lemma 7.2. *If $w \geq 1$ and $n \geq 4$, then $\|\mathcal{I}_3[s^n]\|_{\mathcal{A}} \leq 2^n w^3$ and $\|P_n\|_{\mathcal{A}} \leq w^n + 2^n w^3$.*

Proof. For $n \geq 1$, $\mathcal{I}_3[s^n] = (-1)^n \ell_{-1} + \ell_1 + 2^n \ell_2$. For $w \geq 1$, $\|\ell_{-1}\|_{\mathcal{A}} \leq w^3$, $\|\ell_1\|_{\mathcal{A}} \leq 2w^3$, $\|\ell_2\|_{\mathcal{A}} \leq w^3/3$, and $3 + 2^n/3 \leq 2^n$ for $n \geq 3$. □

Proposition 7.3. *Assume $k \geq 10^9$, $r\theta \leq 1/64$, $w = 4\sqrt{r/\theta}$ and $y := 40w/k < 1$. Then*

$$h = \sum_{n \geq 4} \beta_n P_n \quad \text{in } \mathcal{A}_w, \qquad \|h\|_{\mathcal{A}} \leq 4k\, \frac{y^4}{1-y} + 4k\, w^3\, \frac{(80/k)^4}{1 - 80/k}.$$

Proof. By Lemmas 7.1 and 7.2, the series $\sum_{n \geq 4} \beta_n P_n$ converges absolutely in $\mathcal{A}_w$, and $\sum_n \beta_n \mathcal{I}_3[s^n]$ converges to a cubic polynomial. Hence

$$p := h - \sum_{n \geq 4} \beta_n P_n = \sum_{n \leq 3} \beta_n s^n + \sum_{n \geq 4} \beta_n \mathcal{I}_3[s^n]$$

is a cubic polynomial; it vanishes at the four nodes because h and every P_n do, so $p \equiv 0$. The bound follows by summing $4 \cdot 40^n k^{1-n}(w^n + 2^n w^3)$ over $n \geq 4$. □

The first term is the quartic main term: since $w^2 = 16r/\theta$ and $k\theta \asymp 1$, it is $\asymp k^{-3} w^4 \asymp r^2/k$. For the three-point reference the analogous main term is $\asymp k^{-2} w^3 \asymp (r^3/k)^{1/2}$, which is the source of the cubic wedge (1.2).

8. Proof of the Main Theorem

Lemma 8.1 (Inertia)**.** *Let S be invertible, J a signature matrix, and $T = S(J + F)S^T$ with F symmetric and $\|F\| < 1$. Then T and J have the same inertia.*

Proof. For $\lambda \in [0, 1]$, $\sigma_{\min}(J + \lambda F) \geq 1 - \lambda\|F\| > 0$, so the eigenvalues of the symmetric family $J + \lambda F$ never cross 0; hence $J + F$ has the inertia of J, and Sylvester's law of inertia transfers it to T. □

Proof of the Main Theorem. Let $r \geq 2$ and $k \geq 10^{11} r^2$; then $k \geq 4 \cdot 10^{11}$, so Inputs A and B and all lemmas above apply. By Lemma 4.3, $r\theta < 2.002\, r/k \leq 2.002/(10^{11} r)$, which is $\leq 1/64$ and $< \log 1.45 < \log \alpha$; so Lemma 4.4, (5.1), Corollary 6.7 and Proposition 7.3 hold. Since $k\theta > 0.53$, $w^2 = 16r/\theta < 16rk/0.53$ and

$$y^2 = \frac{1600\, w^2}{k^2} < \frac{25600}{0.53} \cdot \frac{r}{k} \leq \frac{25600}{0.53 \cdot 2 \cdot 10^{11}} < 2.5 \cdot 10^{-7}.$$

Hence $4ky^4/(1-y) < 4\,(25600/0.53)^2\, 10^{-11}/(1-y) < 0.09337$, and the second term in Proposition 7.3 equals $4 \cdot 80^4 (y/40)^3/(1 - 80/k) < 3.1 \cdot 10^{-7}$. Therefore $\|h\|_{\mathcal{A}} < 0.0934$ and $\|e^h - 1\|_{\mathcal{A}} < e^{0.0934} - 1 < 0.098$ (Appendix A).

Let $H' = [b_{d-i-j}]_{i,j=0}^{d}$. Since $b_s = c_s e^{h(s)}$ at the integers $|s| \leq d$, $H' - M = M \circ (e^h - 1)$, and Corollary 6.7 gives $\|S^{-1}(H' - M)S^{-T}\| \leq \|e^h - 1\|_{\mathcal{A}} < 1$. By (5.1) and Lemma 8.1, H' has the inertia of J; J has $\lfloor r/2 \rfloor$ negative entries, so $\operatorname{sgn} \det H' = (-1)^{\lfloor r/2 \rfloor}$.

Finally, $a_{k+d-i-j} = a_k \rho^d\, \rho^{-i} \rho^{-j}\, b_{d-i-j}$ (all indices are $\geq k - d > 0$), so $H = [a_{k+d-i-j}]$ is a positive diagonal congruence of a positive multiple of H', and $\operatorname{sgn} \det H = (-1)^{\lfloor r/2 \rfloor}$. Reversing the order of the columns of $[a_{k+j-i}]$ gives H, at the cost of the sign of the reversal permutation, $(-1)^{\lfloor r/2 \rfloor}$. Hence $D_{r,k} = (-1)^{\lfloor r/2 \rfloor} \det H > 0$. □

Remark 8.2 (The constant). The main term is at most $4(25600/0.53)^2/C$ when $k \geq Cr^2$, so the argument closes as soon as this is $< \log 2$, i.e. for every $C > 1.35 \cdot 10^{10}$. Using the cruder windows $k\theta > 0.4$, $\alpha > 4/3$ one gets $C > 2.37 \cdot 10^{10}$. No effort has been made to optimize Input A, which is the dominant loss (the true coefficients satisfy $|f^{(n)}(k)/n!|\, k^{n-1} < 0.3$, cf. Table 2, versus $3 \cdot 40^n$).

9. Numerical illustration

Nothing in this section is used in the proof. We use the certified values of λ_n, $n \leq 12016$, published with [9] (relative width $< 3 \cdot 10^{-91}$), which we cross-checked against our own quadrature of (1.1). At $k = 12000$ one finds $k\tau_k = 1.74995$, $k^2|\tau_k - \mathcal{T}(k)| = 1.017$ (compare 20 in Input B), $k^2 \Delta_k = 1.7225$, $\alpha = 1.7777$, $k\theta = 0.7655$.

k	r	three-point	four-point	ratio	four-point$\cdot k/r^2$
11990	3	$4.86 \cdot 10^{-3}$	$1.77 \cdot 10^{-5}$	275	0.024
11990	6	$4.00 \cdot 10^{-2}$	$2.17 \cdot 10^{-4}$	184	0.072
11990	8	$7.93 \cdot 10^{-2}$	$5.02 \cdot 10^{-4}$	158	0.094
11990	12	$1.88 \cdot 10^{-1}$	$1.49 \cdot 10^{-3}$	127	0.124
11990	16	$3.31 \cdot 10^{-1}$	$3.06 \cdot 10^{-3}$	108	0.144
3000	8	$1.61 \cdot 10^{-1}$	$1.93 \cdot 10^{-3}$	83	0.091
6000	8	$1.13 \cdot 10^{-1}$	$9.86 \cdot 10^{-4}$	114	0.092

Table 1: Actual whitened perturbation $\|S^{-1}(H' - M)S^{-T}\|$ on the true ξ-coefficients, for the three-point reference of [10] and the four-point reference of this paper. At fixed r the four-point error decays like k^{-1} and the three-point error like $k^{-1/2}$.

Remark 9.1 (Outlook). Heuristically, a reference with m exact matching points and an exact LDL^T with q-Pascal factor would give a main term $\asymp k^{1-m} w^m \asymp r^{m/2} k^{1-m/2}$, i.e. a

n	$\lvert f^{(n)}(k)/n!\rvert\, k^{n-1}$	$\lvert[s^n]\ell\rvert\, k^{n-1}$	$\lvert\beta_n\rvert\, k^{n-1}$	bound $4 \cdot 40^n$
3	0.287	0.287	$9.0 \cdot 10^{-6}$	$2.6 \cdot 10^5$
4	0.142	0.196	0.054	$1.0 \cdot 10^7$
5	0.085	0.157	0.072	$4.1 \cdot 10^8$
6	0.056	0.139	0.083	$1.6 \cdot 10^{10}$
8	0.030	0.132	0.102	$2.6 \cdot 10^{13}$

Table 2: Taylor coefficients at $k = 12000$, computed by Cauchy integrals of $\log I(z) - \log\Gamma(2z+1)$ and of (4.1). The predicted size of the dominant term $\beta_4\mathcal{R}(P_4)$ at $(k, r) = (11990, 8)$ is $5.0 \cdot 10^{-4}$, in agreement with Table 1.

wedge $k \gtrsim r^{m/(m-2)}$. Whether such references exist for $m \geq 5$ (for instance among ${}_r\phi_s$-type sequences) is open; in any case this route cannot reach the critical regime $k = O(r)$, where w becomes comparable to the radius of analyticity.

A. Certified scalar inequalities and reproducibility

All scalar inequalities in Sections 3–8 are verified by the ancillary script `certificate.py` using Arb ball arithmetic [7] at 256 bits. Each window has the form $c(k)/k$ or $c(k)/k^2$ with c monotone in the favourable direction, so evaluation at $k = 12000$ suffices. The certified values are:

quantity	certified enclosure	used
$k\tau_k$	$[1.712619,\ 2.001667]$	(1.71, 2.002)
$k^2\Delta_k$	$[1.675686,\ 2.003334]$	(1.675, 2.004)
α	$[1.458989,\ 2.392839]$	(1.45, 2.40)
$k\theta$	$[0.537774,\ 2.002]$	(0.53, 2.002)
$k\kappa$	≤ 6.451	< 8.4
y (wedge)	$\leq 4.915 \cdot 10^{-4}$	
$\lVert h\rVert_{\mathcal{A}}$ (wedge)	≤ 0.0933691	< 0.0934
$\lVert e^h - 1\rVert_{\mathcal{A}}$	< 0.098	< 1

Further ancillary scripts (not needed for the proof) check: `sym_check.py` — symbolic verification (generic parameters) of Lemma 4.2, the integer cocycle, Lemma 5.1, Proposition 5.2 for $i, j < 7$, Lemmas 6.1–6.2 and Lemma 7.2; `gateB_numeric.py`, `response_scaling.py` — the response bound of Proposition 6.5, computed from a numerical LDL^T of M at 260 digits; `xi_data.py`, `whitened_experiment.py`, `taylor_check.py` — Section 9.

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